

Practical Privacy-Preserving Behaviour-Based Authentication on the Edge

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ABSTRACT

Sensing-enabled Internet of Things (IoT) devices provide an attractive source of data that can be used for seamless authentication of people located within the reach of these sensors. However, current behaviour-biometric solutions often require that the data is centrally stored and processed. This may pose privacy risks, as the data may be used not only for identifying a user, but also for inferring a user's activities within an IoT environment. The risk can be alleviated by shifting the processing to the edge, so that only selected, obfuscated data is sent to the cloud. Previous work proposes a conceptually elegant privacy-preserving distributed deep learning inference method for user identification. In this work we attempt to actually implement such a method distributed over edge devices and the cloud, and uncover significant challenges related to hardware and system limitations of the distributed setup. We then propose amelioration strategies for privacy-preserving distributed deep learning training. Nevertheless, these strategies only partly address the identified issues, calling for further work in bringing distributed sensor-based behaviour authentication to reality.

ACM Reference Format:

Andraž Krašovec and Veljko Pejović. 2022. Practical Privacy-Preserving Behaviour-Based Authentication on the Edge. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3544793.3560356>

1 PRIVACY-PRESERVING BEHAVIOUR-BASED AUTHENTICATION

Behavioral biometrics that rely on non-intrusive data sources, such as sensors distributed throughout the environment, promise to address the deficiencies of password- and biometric-based authentication [4]. Yet, the data used for behavioural-based authentication, for instance, readings from infrared presence sensors, on-device accelerometers, and others, can also be used for inferring non-intended aspects of behaviour, such the activity a user is performing.

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

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ACM ISBN 978-1-4503-9423-9/22/09.

<https://doi.org/10.1145/3544793.3560356>

This issue is increasingly important as third-party authentication services grow in popularity.

Privacy adversarial network (PAN) [5] distributes a deep learning (DL) model over a client and a server, and harnesses adversarial learning to obfuscate the data the server receives, so that the information pertaining to the intended classification task is preserved, yet information pertaining to malicious classification or signal reconstruction is lost. In [3] we propose a PAN-based approach for behaviour-based identification in IoT environments. Our proposed DL model consists of: *i*) the encoder, *ii*) the user identity classifier, and *iii*) the (malicious) activity classifier. The training part consists of three phases: (i) user classifier training, where we jointly train the encoder and the user classifier, maximising the accuracy of user identity inference, *ii*) activity classifier training, where we train the classifier based on the data passed through the encoder, this time maximising the activity inference accuracy, and *iii*) the complete model training, where we utilise improvements in both classifiers to guide the encoder to simultaneously retain the maximum amount of information about user's identity and minimise the amount of information about user's current activity.

2 TOWARDS PRACTICAL IMPLEMENTATION ON THE EDGE

The above distributed authentication concept ensures a full privacy of a user only if the encoder part runs on device(s) that a user has a full control of. Furthermore, the DL model training should be conducted in a distributed manner prevents the server from seeing raw sensor data. Therefore, in this work we implement both the training and the privacy-preserving identity inference in a fully distributed manner over edge devices and a cloud.

2.1 Methodology

We implement the concept in Python taking roughly 1100 lines of code and use PyTorch as a DL framework. Our DL model architecture is distributed over Raspberry Pi 3B (RPI) edge devices and a server. We closely follow the distributed training process explained in the previous section with the addition of primitives for inter-device communication, for which we design a custom application layer protocol built on top of sockets. In a previous experiment we collected the data of 15 users performing three different tasks in an office-like sensor-equipped environment [2]. Each of the tasks is repeated in two sessions during the same day. We use time and frequency domain features, such as mean, standard deviation, mean

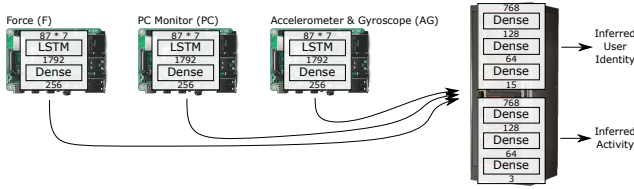


Figure 1: The final architecture of the privacy-preserving authentication system (the initial version joins the three encoders on a single RPi device). Three RPis are gathering data from their respective sensors and training their individual parts of the encoder network. User- and activity- classifiers are in the cloud and perform the inference based on the obfuscated information received from RPis.

crossing rate, and power spectrum, calculated on one second time segments of the data from an accelerometer, gyroscope, four force sensors, and PC usage statistics. We use the first sessions’ data for training, the second sessions’ data for identity and task (i.e. activity) inference.

2.2 Preliminary results and challenges

Our goal is to have a practical prototype system that would: *i*) authenticate a user¹, *ii*) lead to poor activity inference at a server, *iii*) complete the distributed training within 16 hours (representing time between two working days).

We first compare the training time of a distributed setup and the centralized training of the network on the server. We observe that the centralised training completes within less than an hour and achieves 79% user identity classification (c.f. 7% baseline), while successfully preventing activity (i.e. task type) inference which, with 34% accuracy, is no better than random guessing. The distributed training, however, does not converge within the allocated time. To speed the training up, we harness **feature replay** [1]. While training neural networks, traditionally higher layers have to wait to receive gradients of previous layers. In heterogeneous setups this may incur a significant delay. Thus, to break the “backpropagation lock”, we replay data from previous mini batches to update the model parameters on a given device. More specifically: the first mini batch passes through the encoder on a RPi and the output is sent to the cloud; the RPi remembers the mini batch that has just passed through the encoder; the first mini batch passes through the user- and/or activity- classifier, the loss function is calculated, and the backpropagation algorithm gets executed on the cloud, at the same time, the second mini batch passes through the RPi; the RPi receives gradients from the cloud, replays the data of *the first mini batch* through the encoder and, using the gradients, completes the first backpropagation pass. Repeating this process accelerates the learning, yet incurs certain inaccuracies, as stale information is used for training.

Despite the inclusion of feature replay, the training time for a single epoch at RPi takes almost two orders of magnitude more time

¹In [2] we show that a user’s behavior is not time-invariant, thus highly accurate identification (e.g. >95% acc.) is possible only within the same session, i.e. in case of continuous authentication, not across sessions as is the case in our experiments, where the accuracy of 70% is deemed acceptable.

that training on the server – 19.2 min c.f. 6.7 s. Thus, we **distribute the load further** by splitting the encoder into three separate entities for each type of data (acceleration and gyration, force, and PC usage), each hosted on a separate RPi device (as shown in Figure 1). Unfortunately, such a setup achieves a user inference accuracy of only 55%. We believe the reason lies in the interdependence of data from different sensor modalities, which is lost in the case of obfuscating the data separately and then combining them further down in the DL model.

Not all data is equally important, thus, we devise an **input data selection** method similar to boosting-based classification. With it, we try to decrease per-epoch training time, as well as number of epoch needed for successful training. In each epoch, we select a predetermined number of cases of each user, based on the achieved per-user accuracy in the previous epoch. The lower the accuracy, the higher the number of samples taken for a given user. We observe a drop in per-epoch time to 3.25 seconds (centralised training). However, the number of epochs needed to achieve accuracy results comparable to our previous method increases. In the same amount of time allocated for training, using data selection, we observe a 2% decrease in accuracy, while destabilising the training process.

Finally, trading some accuracy for faster running time, we implement a **quantisation-aware training** procedure. Unfortunately, at the time of writing, PyTorch does not enable us to quantise the LSTM layer of the network. As this is the most computationally intensive part of the network, we observe negligible improvement in running times.

3 CONCLUSION

In this work we built a privacy-preserving user authentication system and analysed the practicality of such an approach, relying on RPi as our edge device(s). Our study indicates prohibitively slow training of a distributed privacy-preserving DL model on (even relatively powerful) edge devices. Using feature replay, load distribution, input data selection, and quantization we attempt to ameliorate these issues, but without major progress. While the issues stem from the inherent hardware limitations, such as insufficient RAM and the lack of GPUs, efforts on improving the efficiency of low-level primitives for edge DL could bring us closer to a practical implementation of an IoT-based authentication system.

REFERENCES

- [1] Zhouyuan Huo, Bin Gu, and Heng Huang. 2018. Training Neural Networks Using Features Replay. In *Advances in Neural Information Processing Systems (NIPS 2018, Vol. 31)*. 6659–6668.
- [2] Andraž Krašovec, Daniel Pellarini, Dimitrios Geneiatakis, Gianmarco Baldini, and Veljko Pejović. 2020. Not quite yourself today: Behaviour-based continuous authentication in IoT environments. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–29.
- [3] Andraž Krašovec, Gianmarco Baldini, and Veljko Pejović. 2021. Opposing Data Exploitation: Behaviour Biometrics for Privacy-Preserving Authentication in IoT Environments. In *The 16th International Conference on Availability, Reliability and Security (ARES 2021)*. Article 117.
- [4] Yunji Liang, Sagar Samtani, Bin Guo, and Zhiwen Yu. 2020. Behavioral Biometrics for Continuous Authentication in the Internet-of-Things Era: An Artificial Intelligence Perspective. *IEEE Internet of Things Journal* 7, 9 (2020), 9128–9143.
- [5] Sicong Liu, Junzhao Du, Anshumali Shrivastava, and Lin Zhong. 2019. Privacy adversarial network: Representation learning for mobile data privacy. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 4 (2019).